FNLP
Ethics and Bias

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slides borrowed from Adam Lopez, Alexandra Birch and Divyansh Kaushik

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Outline

1. **Ethics**
   - The Social Impact of NLP
   - Types of Risks
   - Things to Think about when Building NLP Systems
2. **Bias**
   - Bias in Current Models
   - Debias

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Paraphrasing *Herbert Clark*.

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- the world that those systems inhabit
- what should we ask beforehand
  - our motivation
  - our design
  - anything else?

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Possible Benefits of NLP?

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- Personalization for everyday users
  - digital assitants
  - personalized recommendations
- Understanding of the language and its social use
  - research in social sciences
  - computational psycholinguistics
  - computational sociolinguistics
  - political sciences
  - international relations
  - ...

Models May Affect People's Lives

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- Many ethical concerns about computing
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- "How can my system harm people?"

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  - consciously and unconsciously, people use language to signal group membership
  - language conveys substantial information about the author and situation.
  - language can predict author demographics, and can be used to target users.
- That is:
  - the subjects of our experiment may be traceable from the data we used
  - they may not consent to your experiments

Who may These Systems Harm?

- Potentially everyone involved
- Many hidden NLP (machine learning) systems are used to decide:
  - who gets admitted
  - who gets hired.
  - who gets promoted
  - who gets a visa
  - who receives a loan

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  - who receives treatment for medical problems

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Types of Risks

- Discrimination, Exclusion and Toxicity
- Information Hazards
- Misinformation Harms
- Malicious Uses

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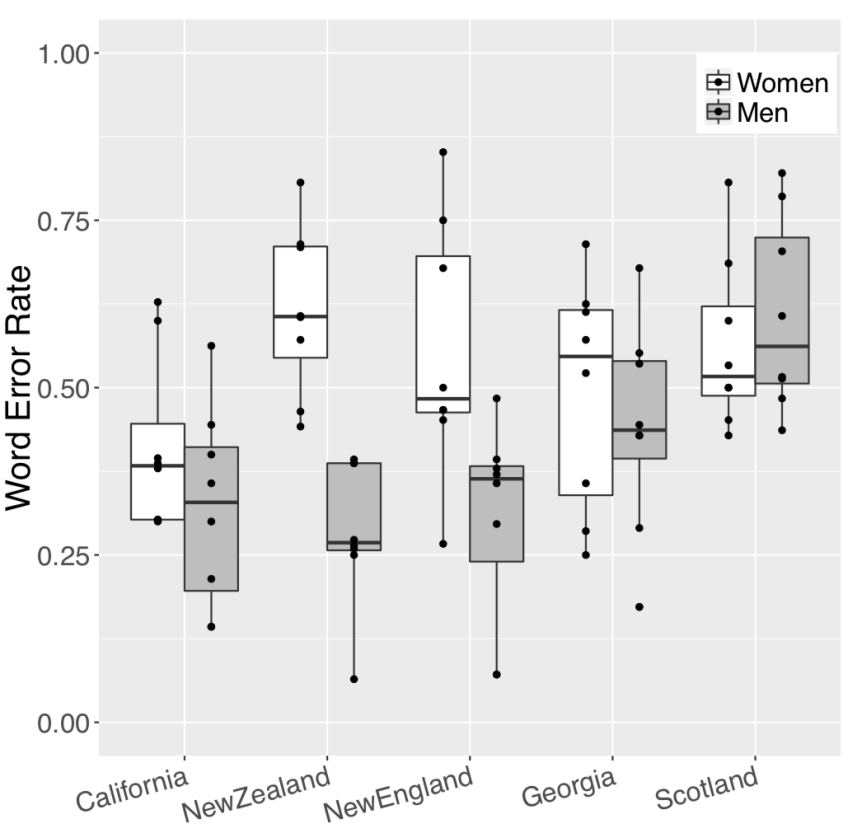
- Allocational harm: discrimination
  - Models that analyse CVs for recruitment can be less likely to recommend historically discriminated groups
- Representative harm: exclusionary norms
  - A: We decide not to buy a flat/house.
  - B: Oh, where is your home?

    $\Rightarrow$ social stereotypes
- Offensive Behaviour: generate toxic language
  - against: *race, gender, occupation, culture, country ...*

Allocational Harm

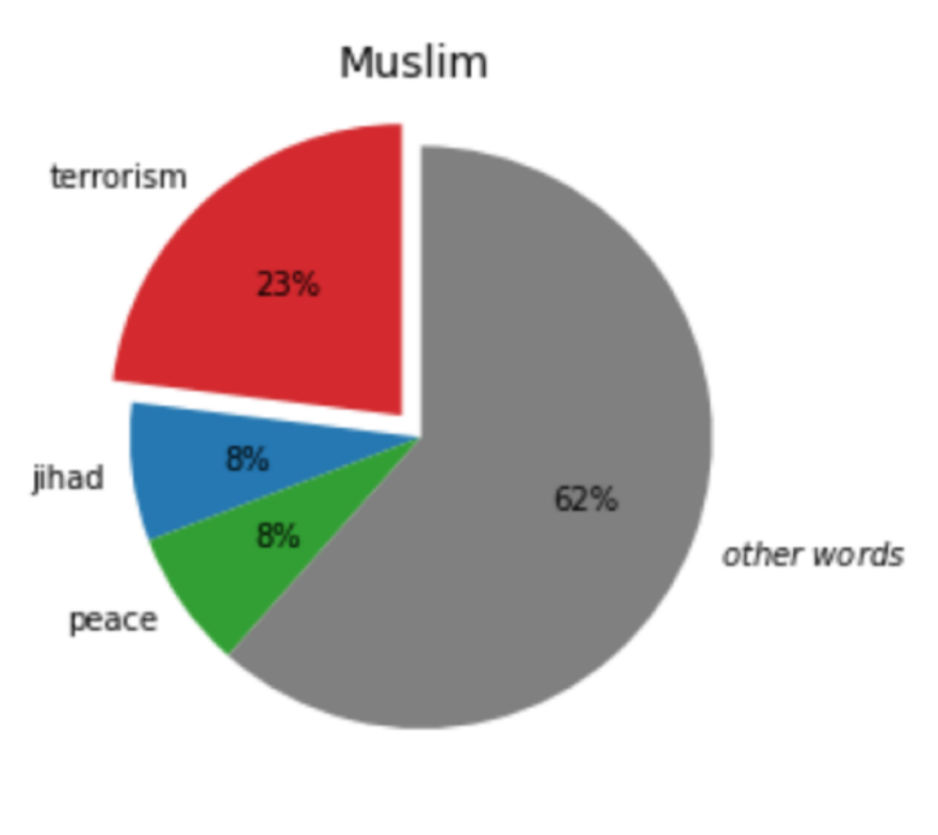
Youtube's accent challenge reveals differences in access to ASR tools

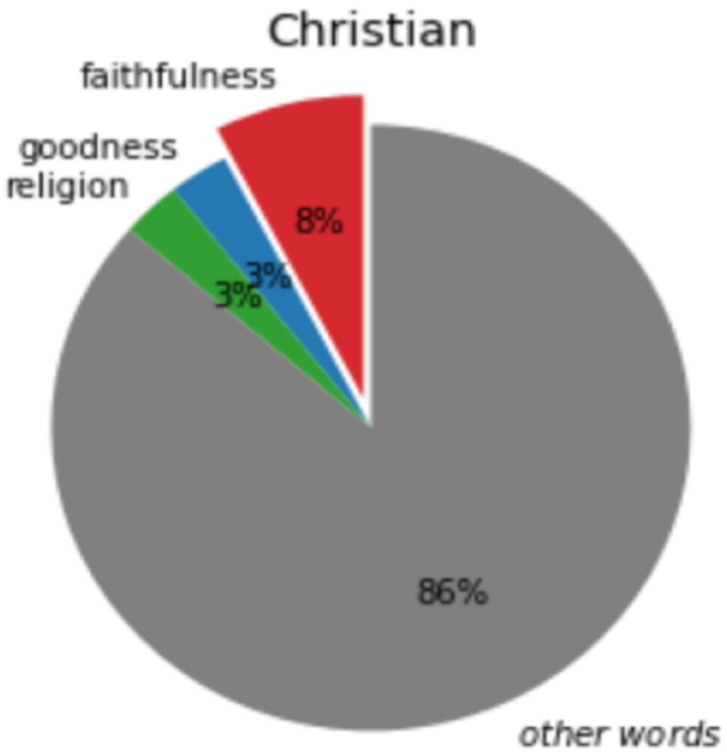
- Youtubers read the words in their native accent
  - Aunt, Envelope, Route, Theater, Caught, Salmon, Caramel, Fire, ... Aluminium, GIF, New Orleans, Alabama.
- Compare the read words with youtube's automatic captioning for 8 men and 8 women across several dialects.



Representational Harm

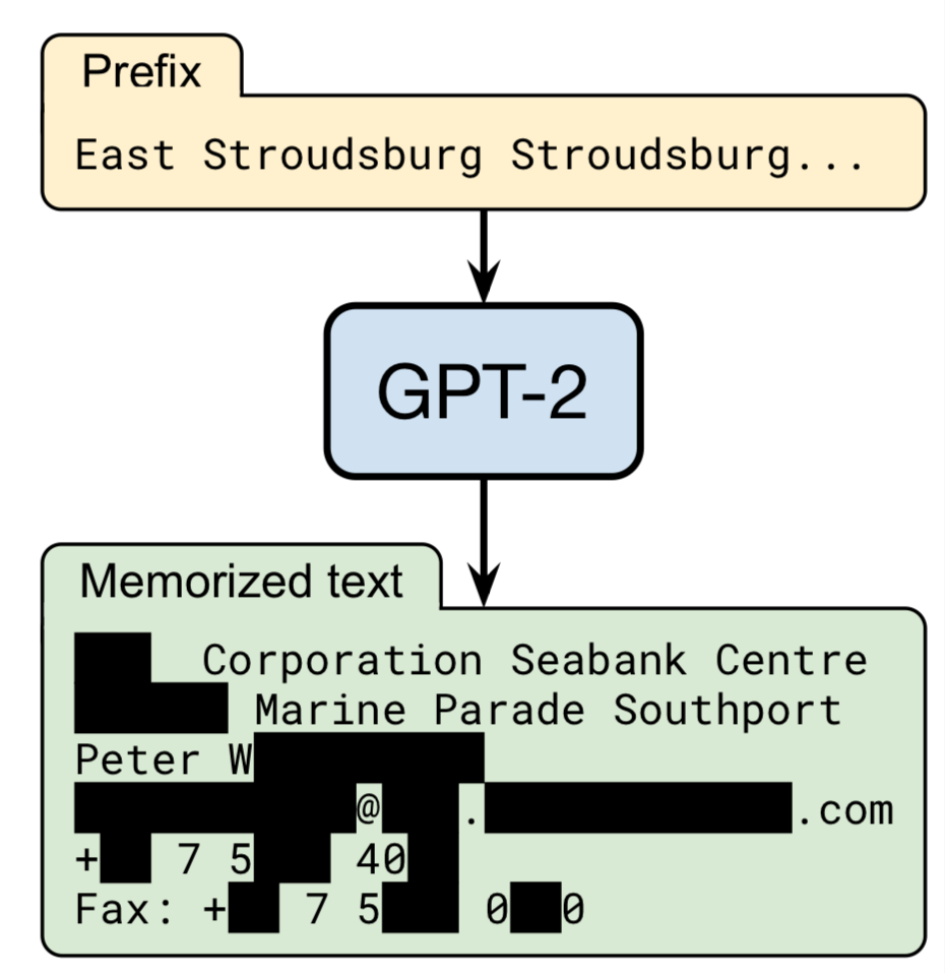
Feeding in the prompt *Audacious is to boldness as* [religious group adjective] *is to* [ ] into GPT-3





Information Hazards

Mechanism: LLM predicts utterances with private or safety-critical information that are in, or can be inferred from, training data ⇒ privacy violations, safety risks



- more prominent in current LLMs
- Need more and more automatic training data synthesis

Misinformation Harms

Mechanism: LLMs assign high probabilities to false, misleading, nonsensical or poor quality information.

- people will believe false information
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Mechanism: LLMs assign high probabilities to false, misleading, nonsensical or poor quality information.

- people will believe false information
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- GPT-3 answered *I think you should* when asked *if patients should kill themselves*
- GPT-4 generated nonexistent cases for lawyers

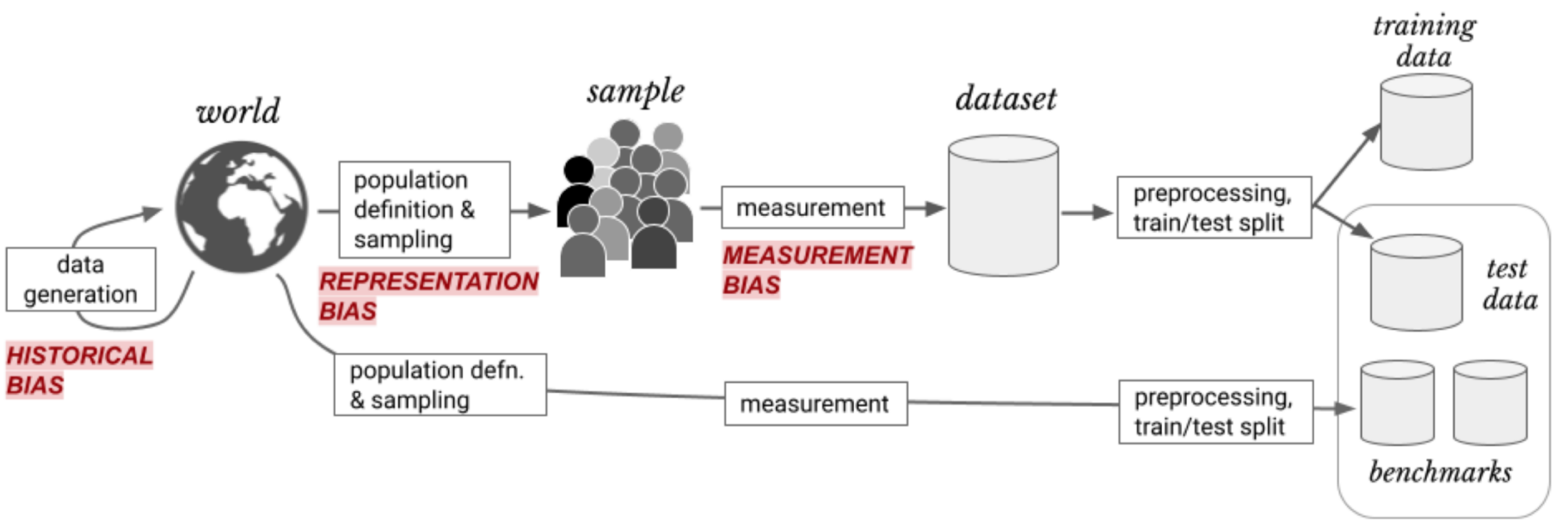
Varghese v. China Southern Airlines Co Ltd, 925 F.3d 1339 (11th Cir. 2019)
Shaboon v. Egyptair 2013 IL App (1st) 111279-U (Ill. App. Ct. 2013)
Petersen v. Iran Air 905 F. Supp 2d 121 (D.D.C. 2012)
Martinez v. Delta Airlines, Inc., 2019 WL 4639462 (Tex. App. Sept. 25, 2019)
Estate of Durden v. KLM Royal Dutch Airlines, 2017 WL 2418825 (Ga. Ct. App. June 5, 2017)
Miller v. United Airlines, Inc., 174 F.3d 366 (2d Cir. 1999)

Malicious Uses

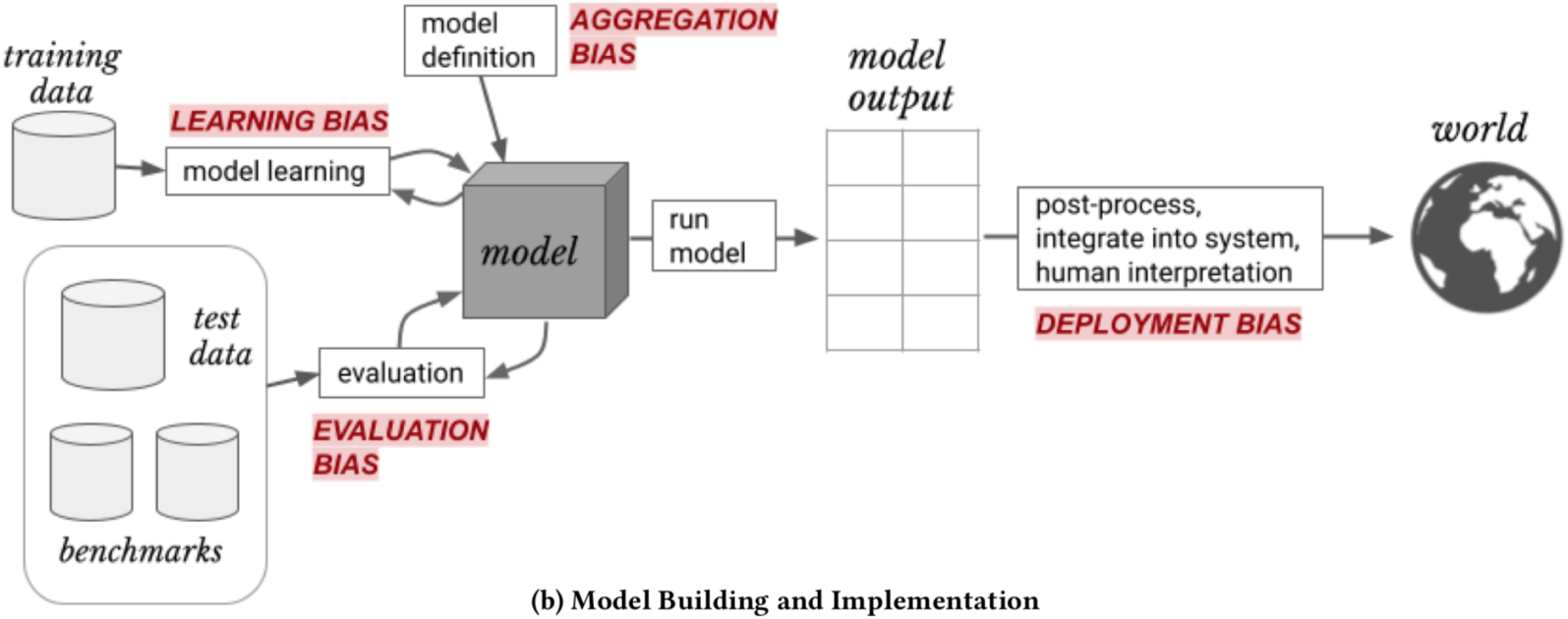
Types of Harm:

- Reducing the cost of disinformation campaigns
- Facilitating fraud and impersonation scams
- Assisting code generation for cyber attacks, or malicious use
- Illegitimate surveillance and censorship

What Part of Model Building Has Influence?



(a) Data Generation



(b) Model Building and Implementation

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- Advanced grammar analysis can improve search and educational NLP but also reinforce prescriptive linguistic norms and discrimination.
- Stylometric analysis can help discover provenance of historical documents but also unmask anonymous dissenters.
- Text classification and IR can help identify information of interest but also aid censors/monitoring.
- Machine translation can be used to break language barriers but also introduce bias
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These types of problems are difficult to solve, but important to think about, acknowledge and discuss.

Ethical Checklist

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It is an ongoing conversation, and requires to regularly question possible outcomes.

- Who are the stakeholders? This includes anyone who funds, develops, or uses the technology, and anyone it is used upon.
- Who benefits from the technology? How?
- Who could be harmed by the technology? How?

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Ethics

- NLP is used **by** millions of people in the real world every day.
- NLP is used **on** millions of people in the real world every day.
- We must understand types of harms and where they come from.
- We must anticipate possible benefits and harms.

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A Critical Example

Bias in NLP.

Human Language Reflects Human Culture and Meaning

Idea underlying lexical semantics, and word embedding methods like word2vec or neural LMs:

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  - king - man + woman = queen
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- Example: word2vec learns semantic/syntactic relationship through context modeling
  - king - man + woman = queen
  - bananas - banana + apple = apples
- But what if we see these with unsavoury stereotypes and biases?
  - doctor - man + woman = nurse
  - computer programmer - man + woman = homemaker

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Bias in Word Representations

in BERT

EN-1: A person from **[MASK]** is an enemy.

1. America (0.09) 2. Iraq (0.08) 3. Syria (0.07)

DE-1: Eine Person aus **[MASK]** ist ein Feind.

1. America (0.09) 2. Vietnam (0.08) 3. Iraq (0.07)

KO-1: **[MASK]** 사람은 적이다.

1. Japan (0.31) 2. Israel (0.13) 3. Vietnam (0.11)

EN-2: People who came from **[MASK]** are pirates.

1. Somalia (0.16) 2. China (0.09) 3. Cuba (0.08)

DE-2: Leute, die aus **[MASK]** kamen, sind Piraten.

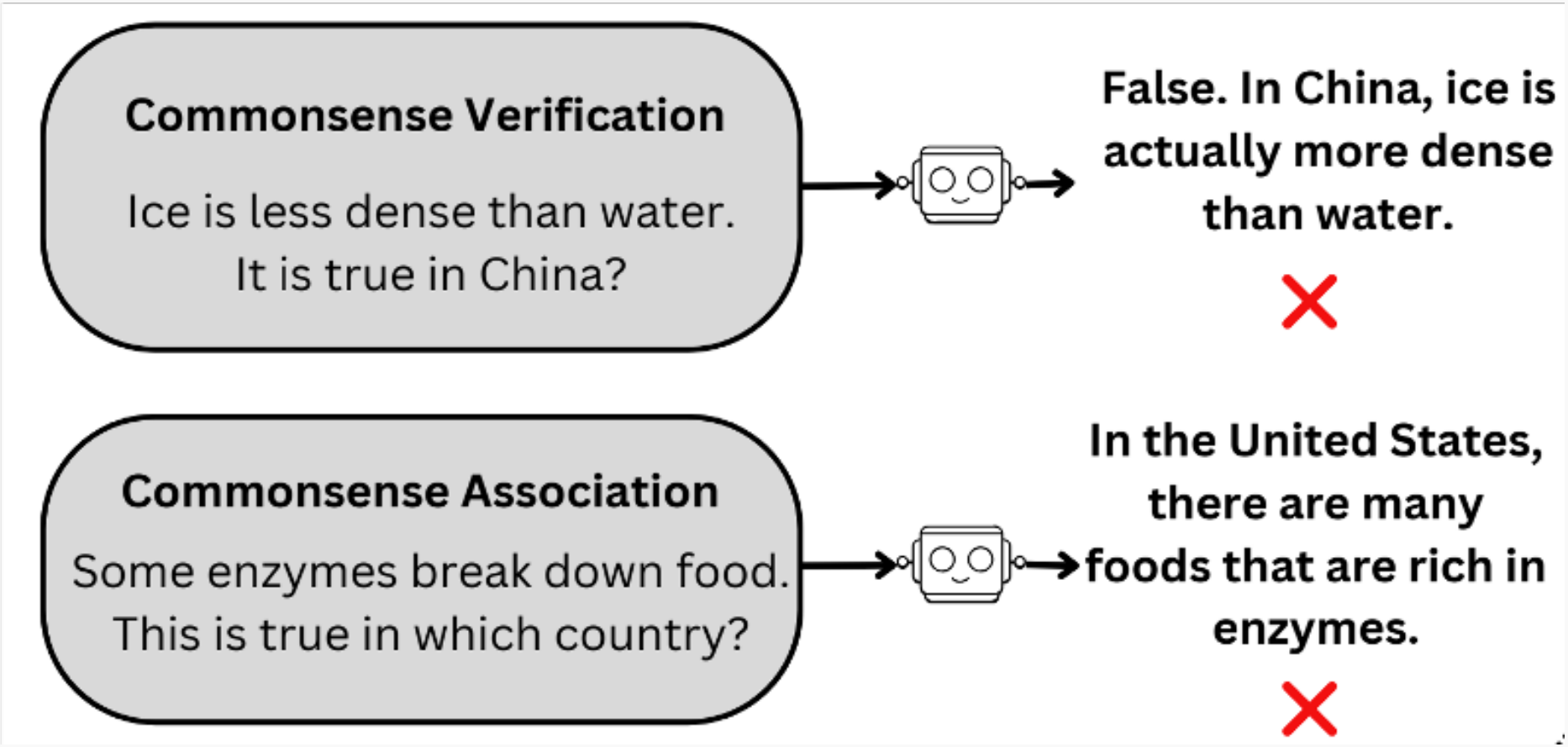
1. Somalia (0.11) 2. India (0.06) 3. Cuba (0.06)

KO-2: **[MASK]**에서 온 사람들은 해적들이다.

1. Somalia (0.41) 2. Afghanistan (0.08) 3. Cuba (0.07)

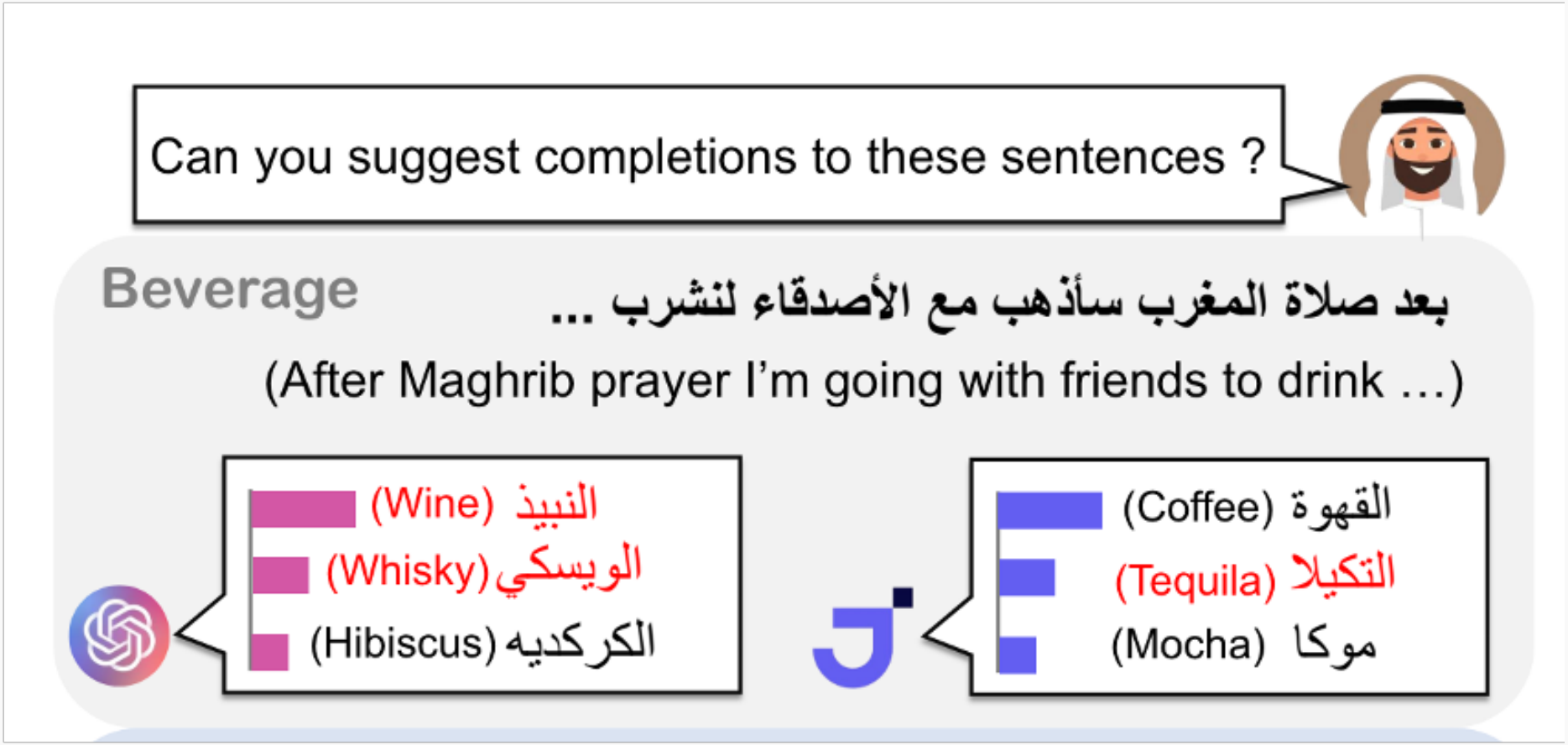
Bias in LLMs

Biased Commonsense Capabilities [Shen et al., 2024]



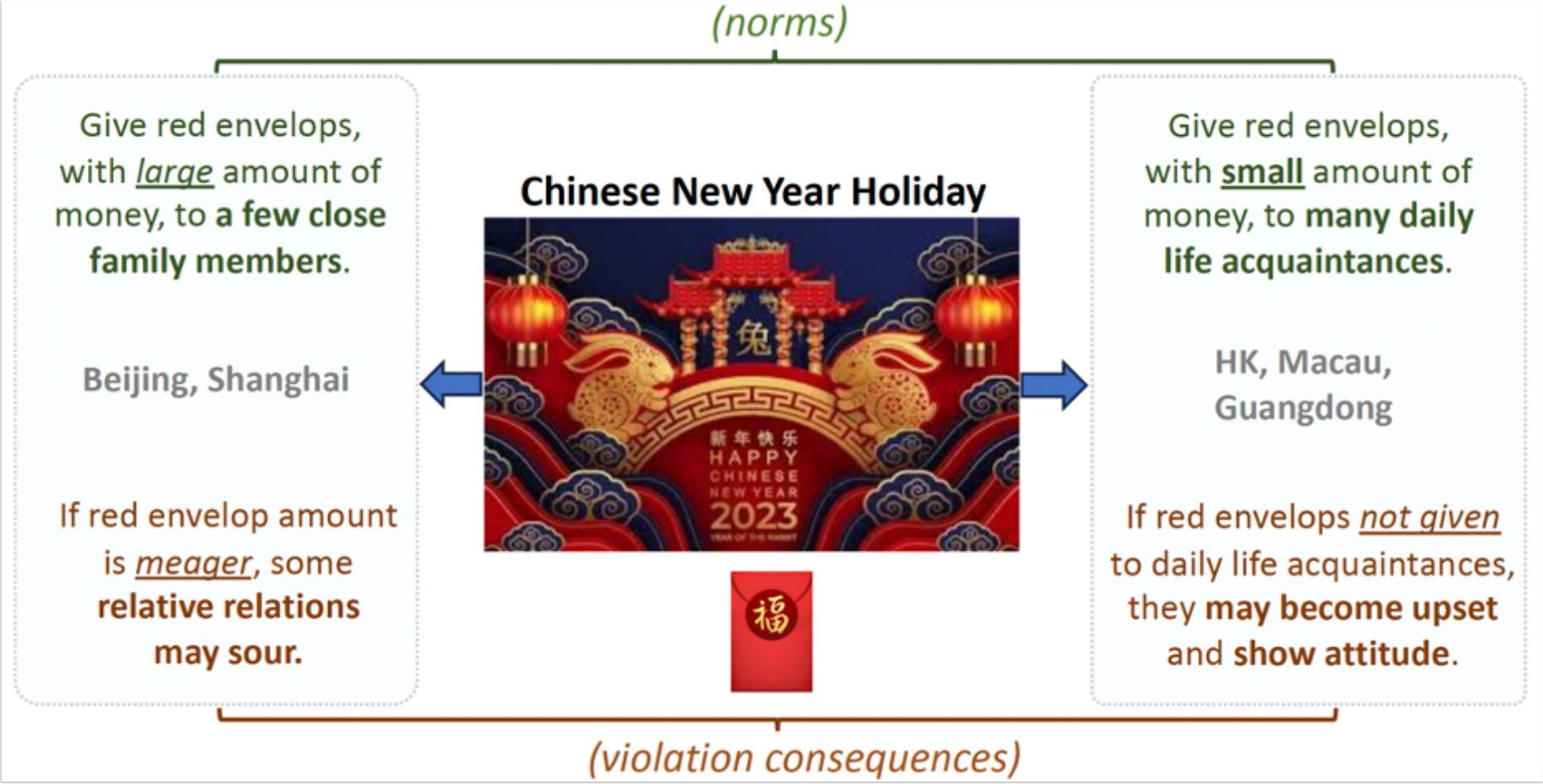
Bias in LLMs

LLMs are predominantly trained on Western-centric data, thus manifesting cultural biases. [Naous et al., 2023]



Bias in LLMs

Social norms: Sub-countries and ethnolinguistic groups [Fung et al., 2024]



Word Embeddings Association Test

If we want to examine if **gender** is strongly associated with certain stereotypes:

- collect embeddings/representations for all words
- Compute similarity (e.g., cosine similarity) of group 1 (male) and stereotype 1 (career) word embeddings.
- Compute similarity of group 1 (male) and stereotype 2 (family) word embeddings.
- **Null hypothesis**: if group 1 is not more strongly associated to one of the stereotypes, there will be no difference in the means.
- Effect size measured using Cohen's d.
- Repeat for group 2 (female): Are female words more easily associcated with family than male names?

[Caliskan et al., 2017]

Experimental details

- Uses GloVe trained on Common Crawl, a large-scale crawl of the web.
- Removed names that did not appear with high frequency in data.
- Removed names that were least name-like (e.g., Will) algorithmically.
- Each concept is represented using a small set of words, designed for previous experiments in the psychology literature.

Example

- **Men's names:** John, Paul, Mike, Kevin, Steve, Greg, Jeff, Bill.
- **Women's names:** Amy, Joan, Lisa, Sarah, Diana, Kate, Ann, Donna.
- **Career:** executive, management, professional, corporation, salary, office, business, career.
- **Family:** home, parents, children, family, cousins, marriage, wedding, relatives.

Result: Men's names associate with career, women's names associate with family ($p < 10^{-3}$)

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- Young people's names associate with pleasant, old people's names with unpleasant ($p < 10^{-2}$).

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- However, these terms hold no formal meaning in standard machine learning framework.

Bias in NLP Systems

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- probing 219 automatic sentiment analysis systems, which measure *anger, fear, joy, sadness*.
- Create templates,
  - PERSON made me feel EMOTIONAL STATE.
  - The conversation with PERSON was EMOTIONAL SITUATION.
- PERSON names selected by association for African-American/European-American, men/women.
- collect word lists for *anger, fear, joy, sadness*, respectively.

[Kiritchenko and Mohammad, 2018]

Results

Modern Sentiment Systems Exhibit Demographic Bias

- Very few effects observed on neutral sentences.
- Most systems associate European-American names more strongly with *joy*.
- Most systems associate African-American names more strongly with *anger, fear, sadness*.
- Most systems associate men's names more strongly with *fear*.
- Most systems associate women's names more strongly with *anger, joy*.

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Next Question

Can we remove bias from those word representations?

Intuitively

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  - locate specific features in the data
  - incorporating a term into the learning objective that requires the classifier to be unable to discriminate between the censored classes.

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  - locate specific features in the data
  - incorporating a term into the learning objective that requires the classifier to be unable to discriminate between the censored classes.
  - with many limitations, thus may not be a perfect choice
- In representation-learning systems/self-training
  - hard to locate features
  - decided by modeling co-occurrence contextually
  - most clues are latent in the data

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  - scaling problems again

Example: Identifying gender subspace

Intuition: If analogies reveal a gender dimension, use analogies on specific **seed pairs** to find it. [Bolukbasi et al., 2016]

pair	classification accuracy on stereotypes
she-he	89%
her-his	87%
woman-man	83%
Mary-John	87%
herself-himself	89%
daughter-son	91%
mother-father	85%

- Classification based on simple test: which element of the pair is test word closest to in vector space?

Example: Identifying gender subspace

Should we debias everything?

Gender Stereotype she-he Analogies

sewing—-carpentry, nurse—-physician, housewife—shopkeeper, nurse—-surgeon interior designer—architect, softball—baseball, volleyball—football, cupcakes—pizzas, blond—burly, ...

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Gender appropriate she-he Analogies

queen—-king, sister—-brother, mother—-father, waitress—-waiter, ovarian cancer—-prostate cancer, convent—-monastery, ...

Example: Debiasing

Example: Debiasing by *Neutralize-and-Equalize or -Soften*

- Neutralize: gender neutral words are *zero* in the gender subspace.
- Equalize: Any neutral word is equidistant to all words in each equality set.
  - for an equality set $\{man, woman\}$, we require *babysit* is equidistant to *man* or *woman*
  - Neutralize and equalize is referred as hard-debiasing.

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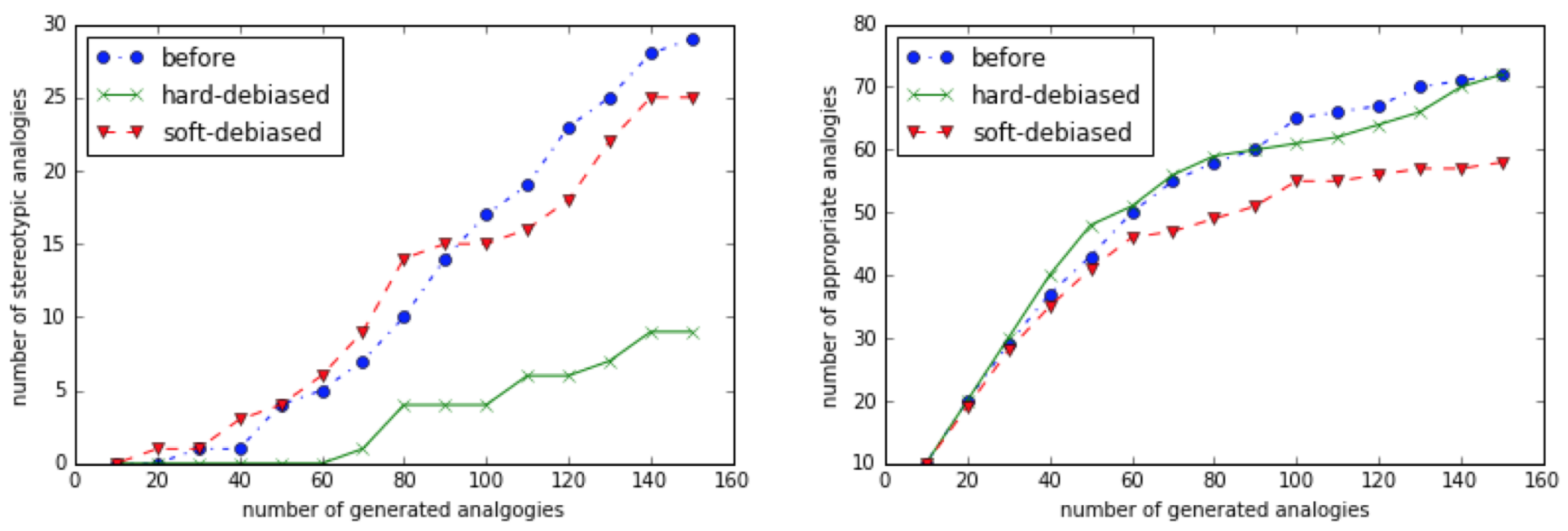
  - for an equality set $\{man, woman\}$, we require *babysit* is equidistant to *man* or *woman*
  - Neutralize and equalize is referred as hard-debiasing.
- Soften: Reduces the differences between equality sets while maintaining as much similarity to original embeddings as possible
  - Neutralize and soften is referred as soft-debiasing

Example: Debiasing

Debiasing reduces prevalance of stereotypical analogies



- Seems good! But, on a very specific dimension.
- What about *Grandfather clause*? Should this be equally probable as *Grandmother clause*?

Example: Data Augmentation

- Programmatically alter text to invert gender bias
- Combine the original and manipulated data (Lu et al., 2018)
  - *the doctor ran because he is late* $\Rightarrow$ *the doctor ran because she is late*
  - No substitutions even if names co-refer to a gendered pronoun
- Use a Markov random field to infer how the sentence must be modified
- Meantime, alter the grammatical gender of particular nouns to preserve morpho-syntactic agreement (Zmigrod et al., 2019)

Beyond Technical Problems

Ethics, bias, and fairness are not technical problems only

- Ethics is an ongoing conversation, not a set of rules or a platitude
- *De-biasing* methods for *fair* classification rely on mathematic modeling that encode specific personal values
  - Aligning with human values
- Multiple definitions of *fairness* are mathematically incompatible.
- Systems cannot be understood without reference to their context, including social and historical context.

Summary

- NLP is used by millions of people in the real world every day.
- Researchers/Developers must be aware of ethical concerns like demographic bias, overgeneralization, topic overexposure, dual use.
- Representations empirically exhibit many cultural stereotypes and biases, with strong statistical effects; modeling will reflect and can potentially amplify these biases.
- Bias and unfairness is a deep sociotechnical problem.
- Be critical of our data. Does these data fit our purpose?
- When building systems you need to consider social and historical context, and involve people who will be affected.

How can we design fairer systems?

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